

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Renee Vincent ASR-192424161 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Renee Vincent ASR

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

Analysis:

- **Session Layer** establishes, manages, and terminates communication sessions between applications. It provides synchronization and dialog control.
- **Data Link Layer** ensures reliable node-to-node communication by framing data, performing error detection, and controlling access to the transmission medium.
- Together, these layers improve communication reliability by maintaining active sessions and ensuring error-free data transfer between adjacent devices.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

Analysis:

- **Transport Layer:** Divides application data into **segments**.
- **Network Layer:** Adds IP addressing, forming **packets**.
- **Data Link Layer:** Encapsulates packets into **frames** by adding MAC addresses and error-checking information.
- **Physical Layer:** Converts frames into **bits** for transmission through the communication medium.

This transformation, known as **encapsulation**, enables efficient and reliable transmission across networks.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

Analysis:

- **MAC Address** identifies devices within the same local network and enables **local delivery**.
- **IP Address** uniquely identifies devices across interconnected networks and enables **global delivery**.
- When data is transmitted, the IP address determines the destination network, while the MAC address delivers the data to the correct device on each local network.
- Together, these addressing methods ensure complete end-to-end communication.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

Analysis:

- The **Data Link Layer** detects transmission errors using techniques such as CRC and checksums.
- The **Transport Layer** ensures reliable delivery through acknowledgments, retransmissions, sequencing, and flow control.
- Together, these layers detect errors early and recover from lost or corrupted data, improving communication reliability.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

Analysis:

- **Application Layer:** Sends HTTP/HTTPS requests.

- **Presentation Layer:** Encrypts/decrypts data (SSL/TLS).
- **Session Layer:** Maintains communication sessions.
- **Transport Layer:** Provides reliable delivery using TCP.
- **Network Layer:** Routes packets using IP addresses.
- **Data Link Layer:** Delivers frames within the local network.
- **Physical Layer:** Transmits bits through cables or wireless media.

Effect of Failure:

- Failure at any layer interrupts communication.
- For example:
 - Physical Layer failure → No network connectivity.
 - Network Layer failure → Packets cannot reach the destination.
 - Transport Layer failure → Data becomes incomplete or lost.
 - Application Layer failure → Web page cannot be accessed.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗ Analyze the problem and explain how the concept map helps in troubleshooting.

Analysis:

- The **Physical Layer** is functioning correctly, indicating cables or wireless signals are operational.

- The **Data Link Layer** is also working, meaning local communication is successful.
- The **Network Layer** has failed, indicating problems such as incorrect IP addressing, routing errors, or router failure.
- Since higher layers depend on the Network Layer, the **Transport** and **Application** layers also fail.
- The concept map helps troubleshoot by identifying the first failed layer, allowing engineers to focus on network configuration or routing instead of checking the physical connection.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20 %	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts ; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25 %	Constructs a clear, logical, and well organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20 %	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20 %	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly

Explanation & Clarity	15 %	Provides a clear, concise, and well structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation
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